• **DRIVEABILITY—ROUGH IDLE, LOW POWER AND/OR STALL AT IDLE WITH DIAGNOSTIC TROUBLE CODES (DTCS) P0300-P0304 STORED IN MEMORY—VEHICLES EQUIPPED WITH 2.0L SPI ENGINE ONLY** • **ELECTRICAL—MALFUNCTION INDICATOR LAMP (MIL) ILLUMINATED—DIAGNOSTIC TROUBLE CODES (DTCS) P0300-P0304 STORED IN MEMORY—VEHICLES EQUIPPED WITH 2.0L SPI ENGINE ONLY**	**Article No. 02-9-1**

FORD: 1997-2002 ESCORT
2000-2002 FOCUS

MERCURY: 1997-1999 TRACER

This article is being republished in its entirety to update the vehicle model lines and model years.

ISSUE

Some vehicles equipped with the 2.0L SPI engine may exhibit a Malfunction Indicator Lamp (MIL) illuminated and Diagnostic Trouble Codes (DTCs) P0300 through P0304 stored in memory. The vehicle may also have a rough idle or reduced power condition, and/or may stall at an idle. This may be caused by sticking exhaust valves due to excessive carbon build-up.

ACTION

Check for faulty fuel injectors and ignition wires. One or more exhaust valves may be temporarily sticking due to carbon build-up between the valve stem and guide. If the diagnostic checks point to a sticking exhaust valve, the cylinder head assembly may need to be replaced with a new level cylinder head assembly that includes revised valve guides and exhaust valves to help prevent carbon build-up. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE
THIS PROCEDURE APPLIES ONLY TO THE 2.0L SPI ENGINE.

This condition usually results in misfire when the engine returns to idle after running over approximately 2000 rpm.

1. Use the Worldwide Diagnostic System (WDS) to test for Power Balance with the engine at idle. Check for cylinder misfire. If misfire is occurring constantly, go to Step 3. If misfire is not occurring, attempt to induce misfire by performing the following step:

   a. Apply the parking brake and keep the transmission in Park or Neutral. Bring the engine to approximately 2800 rpm and hold for 1 minute. Check the WDS Power Balance display while holding engine speed constant. Allow the engine to return to idle and watch the WDS Power Balance display for a misfiring cylinder.

2. If WDS Power Balance indicates a misfiring cylinder with the previous test, one or more exhaust valves may be intermittently sticking due to carbon build-up between the valve stem and guide. Replace the cylinder head and cam assembly with the new level cylinder head and cam assembly built with revised valves and valve guides to help prevent carbon build-up. Consult the appropriate model year Workshop Manual for removal and installation procedures for the Cylinder Head and Cam Assembly.

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3. If WDS Power Balance did not indicate a misfiring cylinder with the previous test or a misfire is occurring constantly, the misfire may not be due to a sticking exhaust valve. Refer to the appropriate model year Powertrain Control/Emissions Diagnosis (PC/ED) Service Manual for further diagnosis. Other possible causes may be, worn or arcing ignition wires, base engine compression, fuel injectors, fuel injector harness resistance or PCM injector driver signal. Perform diagnosis and repair as necessary.
4. After completing either Step 2 or Step 3, replace the PCV Valve (YS4Z-6A666-AA) (Motorcraft EV-244) on model year 2000-2002 Focus or 2000-2002 Escort. Vacuum fluctuations from engine misfire may prematurely wear the PCV valve on those applications.

The following tables list parts that may be required to repair vehicle.

2000-2002 Focus		
Part Description	**Part Number**	**Qty.**
Cylinder Head and Cam Assembly	YS4Z-6049-FA	1
Cylinder Head to Block Gasket	YS4Z-6051-BA	1
Cylinder Head Bolts (pkg of 10)	E4FZ-6065-A	1
Cylinder Head to IMRC Gasket	YS4Z-9439-AA	1
IMRC to Intake Manifold Gasket	YS4Z-9H486-CA	1
Intake Manifold Stud	W706759-S309	2
Exhaust Manifold To Head - Gasket	F6CZ-9448-AC	1
Exhaust Manifold Stud	F7CZ-9454-AA	8
Exhaust Manifold Nut	W701706-S2	8
Spark Plug	AGSF34E-EMF6	4
Exh. Manifold Collector Gasket	F7CZ-9450-FA	1
Catalyst Nut	W703933-S403	4
EGR Tube Gasket	F6CZ-9C435-AA	1
EGR Tube & Exh. Manifold Shield Nut	W520101-S309	7
PCV Valve	YS4Z-6A666-AA	1

2000-2002 Escort		
Part Description	**Part Number**	**Qty.**
Cylinder Head and Cam Assembly	F7CZ-6049-CAA	1
Cylinder Head to Block Gasket	YS4Z-6051-BA	1
Cylinder Head Bolts (pkg of 10)	E4FZ-6065-A	1
Cylinder Head to IMRC Gasket	XS4Z-9439-CA	1
IMRC to Intake Manifold Gasket	XS4Z-9H486-CA	1
Intake Manifold Stud	N808685-S309	2
Exhaust Manifold To Head - Gasket	F6CZ-9448-AC	1
Exhaust Manifold Stud	F7CZ-9454-AA	8
Exhaust Manifold Nut	W701706-S2	8
Spark Plug	AGSF34E-EMF6	4
Water Outlet Gasket	F6CZ-8255-AA	1
Exhaust Manifold To Catalyst Gasket	F7CZ-9450-FA	1
Catalyst Nut	N811499-S101	4
EGR Tube Gasket	F6CZ-9C435-AA	1

1997-1999 Escort/Tracer		
Part Description	**Part Number**	**Qty.**
Cylinder Head and Cam Assembly	F7CZ-6049-CAA	1
Cylinder Head to Block Gasket	F7CZ-6051-BB	1
Cylinder Head Bolts (pkg of 10)	E4FZ-6065-A	1
Cylinder Head to IMRC Gasket	XS4Z-9439-CA	1
IMRC to Intake Manifold Gasket	XS4Z-9H486-CA	1
Intake Manifold Stud	N808685-S309	2
Exhaust Manifold To Head - Gasket	F6CZ-9448-AC	1
Exhaust Manifold Stud	F7CZ-9454-AA	8
Exhaust Manifold Nut	W701706-S2	8
Spark Plug	AGSF34E-EMF6	4
Water Outlet Gasket	F6CZ-8255-AA	1
Exhaust Manifold To Catalyst Gasket	F7CZ-9450-FA	1
Catalyst Nut	N811499-S101	4
EGR Tube Gasket	F6CZ-9C435-AA	1

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 01-4-13
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 499000, 607000, 607400, 607500, 607600, 607700, 608000, 608400, 608500, 608600, 609000, 609400, 609500, 609600, 611000, 611500, 614000, 614500, 614600, 698298